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LC T P Spring symposium: Theoretical Physics and AI

✖ **Factual Error**

Problem 899, CoT Step 5

Model claim: “ $(\gamma^5)^2 = -\mathbb{I}$ ”

Correct fact: γ^5 squares to the identity, $(\gamma^5)^2 = \mathbb{I}$.

Misrecalled property of the chirality matrix propagates through the subsequent projection operator algebra.

⌘ Executional Error

Problem 1199, CoT Step 8

Derivation: Correctly obtains the BPS energy coefficient $C = \frac{1}{4}$.

Code: `C = 1/3` ← **bug**

The analytical reasoning is correct, but the implementation assigns a different numerical value, a transcription error visible only in the code.

Mathematical Error

Problem 993, CoT Step 4

Model writes: $(2x - 1)^2 = x^2 - 2x + 1$

Correct expansion: $(2x - 1)^2 = 4x^2 - 4x + 1$.

The factor of 4 is dropped from the leading coefficient, producing incorrect polynomial coefficients for the Feynman parameter integral.

Error Categories

2

9

⚠ Logical Error

Problem 49, CoT Step 4

Model derives: Amplitude scales as $\mathcal{A}(z) \sim z^2$ at large z .

Model concludes: "...therefore the amplitude scales as z^{-2} ."

The derivation is correct, but the conclusion contradicts it—inverting the sign of the exponent and reversing the large- z behavior.

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Major Error Comparison: Base vs RL vs SFT (10 Overlapping Most-Improved Problems)

